Technical Proposal

Credit Risk Prediction System Using Machine Learning

1. Executive Summary

This proposal outlines the development of a Machine Learning-based Credit Risk Prediction System.
The system aims to classify loan applicants into low-risk and high-risk categories using historical financial data
to reduce default rates and improve decision-making efficiency.

2. Introduction and Background

Financial institutions suffer losses due to loan defaults.
Traditional rule-based systems may fail to detect complex patterns.
Machine Learning enables predictive analytics to improve credit risk assessment accuracy.

3. Proposed Solution & Added Value

- Data preprocessing and cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering
- Model training (Logistic Regression, Random Forest, XGBoost)
- Hyperparameter tuning
- Model evaluation and optimization
- Deployment as REST API

Added Value:
- Reduced default rate
- Faster approval decisions
- Scalable ML-based credit scoring

4. Timeline and Deliverables

Week 1: Requirement Analysis
Week 2: Data Preprocessing & EDA
Week 3: Model Development

Week 4: Model Optimization
Week 5: Deployment (REST API)
Week 6: Final Documentation & Presentation

5. Resources

Human Resources:
- 1 Machine Learning Engineer

Technical Resources:
- Python
- Pandas & NumPy
- Scikit-learn
- XGBoost
- FastAPI
- PostgreSQL (optional)

6. Technical Specifications

- Python 3.10+
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC
- Target ROC-AUC ≥ 0.85
- API response time < 1 second
- Optional Docker containerization

7. Terms and Conditions

- Data confidentiality must be maintained
- Project duration: 6 weeks
- 30 days post-delivery support
- Scope changes require formal approval

8. Conclusion and Call to Action

The proposed ML-based Credit Risk Prediction System offers a scalable and reliable solution for credit risk assessment. We look forward to approval to begin development and deliver a high-performance predictive system.